Sampling and Sample Preparation in Bioanalysis

Ljubica Tasic

1 Introduction

The success of any chemical or physical description of the composition of any sample depends greatly on sampling, sample preparation, and the choice of technique that is picked for such tasks. There are many types of samples, which can have different physical states at room (standard) temperature and pressure, such as gas, liquid and solid, or a mix of those like emulsions, suspensions, or aerosols (Pawliszyn 2010). That is why sampling and its representativity are of crucial importance when aimed to the best possible analytical sample description. It is important to say that there is no perfect technique for all types of samples, nor the best all-in-one analysis, and each sample, even if not of complex nature, must be taken as unique in the best effort to achieve as good as possible sample composition description.

All samples are made from elements, molecules, and their mixtures and some are the major, others the minor or trace components (Clark et al. 2016). Thus, it is important to establish the goals previous to any analysis to be performed and develop a suitable strategy to make as few as possible, and the smallest as possible errors, in defining the chemical and physical compositions of the studied samples (Pawliszyn 2010; Clark et al. 2016; Berndt 2020).

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2 Sampling and Biosafety

The number of samples and the samples' physical state requires the use of suitable tools and sampling (Berndt 2020). Sample must be representative (Nahorniak et al. 2015) and sufficient (Suresh et al. 2011) in terms of number and quantity for the analytical method or methods, which are going to be used in sample analyses. Also, sample handling, transportation, storage, and preservation (Nocerino et al. 2005; Tripathi et al. 2020) must be discussed before any analysis. Any sample contamination must be avoided or reduced to the lowest possible chance. It is very important to plan and design a study to avoid data interpretation in a biased manner. One must think about what are the aims of a study and if the designed protocol is the most appropriate to get the right answer—for example, samples are analyzed to determine their constituents and concentrations (Berndt 2020; Nahorniak et al. 2015; Suresh et al. 2011; Nocerino et al. 2005; Tripathi et al. 2020).

First of all, sampling depends on the sample's physical state and there are some standard procedures used for each sample type that must be followed. Herein, bioanalytical samples are discussed, thus many protocols for biosafety (Dickmann et al. 2015; Flemming 2000) and authorized sampling for research use must also be taken into account previous to any sample collection.

Biosafety is defined as—the maintenance of safe conditions in biological research to prevent harm to workers, non-laboratory organisms, or the environment—(World Health Organisation Biosafety). Not all laboratory settings are equipped to work with human samples, pathogenic bacteria, viruses, and so on. There are four levels of laboratory containment protocol (Kimman et al. 2008) expressed as 1–4 in order of danger, with level 4 posing a high risk to individuals and community, working with pathogens that cause serious human or animal diseases and to spread easily, such as deadly viruses (Ebola virus) for humans. A risk group 1 is assigned (Enserink and Du 2004) to no or low individual and community risk, for example, to work and research that employs microorganisms that are unlikely to cause human and animal diseases (World Health Organisation Biosafety) (Bayot and King 2020).

On the other hand, an authorized sampling includes protocols that follow all ethical and legal rights, and when discussing the use of human biosamples (Bayot and King 2020), they must have the consent of donation. It is also important to respect the legal and ethical rights of the people who donate their samples and to guarantee that they will make a personal contribution to the research undertaken with the collected samples (Singh et al. 2020). When discussing the sampling of many types of biological samples, it is also important to obey the protocols on access to genetic resources and the fair and equitable sharing of benefits arising from their use to the convention on biological diversity (CBD) (Mohammedsaleh and Mohammedsaleh 2014). When legal and ethical rights are obeyed, then it is also important to have in mind that after sample collecting, there are two types of samples to be prepared and kept, a laboratory size sample and a sample for the biobank (Fig. 1).

The following sections bring some important protocols for sampling in regard to physical state and type of samples.

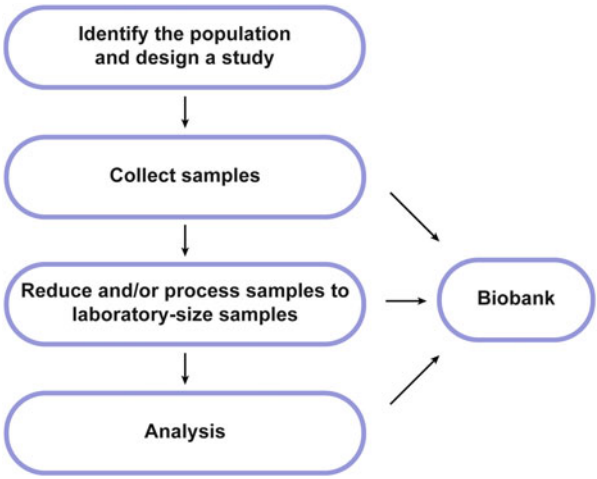


Fig. 1 Illustration of the main steps in a study when discussing sampling in bioanalytics. It is first to define aims and design a study, collect samples and prepare two types of samples from the same cohort—one that is laboratory size and the other to be kept in a biobank

3 Sample Size, Composition, and Errors

The sample size is one of the main issues when biological samples are analyzed. For example, when it is a large population researched, getting a sample from everyone is not possible. Therefore, it is important to take a random sample of individuals or species, which represents the population as a whole. An accurate, and statistically significant result, which leads to a successful conclusion of the study, strongly depends on the sample size. If the sample size is too small, it may include a disproportionate number of individuals who are anomalies or outliers, therefore, the obtained data may lead to a biased assessment of the entire population. But if too big, the whole study becomes complex, expensive, long, and even if it brings accurate results, the benefits of such a study are not justified.

Some tips might help to design the study properly concerning how many samples must be used throughout the research, which depends on the size of the population, margin of error, confidence level, and standard deviation to estimate an ideal, or at least a preferable sample size. For example, a preferable sample size for DNA barcoding is at least 20 samples from the same species. Because of the technical and biological variances among samples, there is a need to calculate the sample size and consider: (1) the technique-based basis, (2) laboratory-based level, (3) aims of the study (disease), and (4) the sample type, before deciding how to proceed. Throughout this chapter, and when important, the sample size is discussed, like the number of individuals, or the quantity of the sample in volume or mass units. And when dealing with the mass or volume units, then, it is important to discuss the sample constituents.

Major elements of a biological sample or a mixture are usually considered to be above 1%, while minor elements make between 0.1 and 1% of a sample, and trace elements are below 0.1% and usually measured in ppm. Trace components go from

1 part per billion (ppb) to 100 parts per million (ppm), but there are also ultra-trace components that account for less than 1 ppb. So, there are some concerns about what one intends to measure. When the sample size is taken into account, its components can be considered as a macro if their mass is above 0.1 g, semi-micro if it is made from 0.01 to 0.1 g, micro components are considered those present in the range 0.0001–0.01 g, and ultra-micro if they make up less than 0.0001 g of a sample.

Analytical determination of very small amounts of substances less than 0.01% and going as low as 10^{-8}% of the sample are called trace analysis and concentration of those substances are expressed in parts per million (ppm) or smaller parts. There are some particular difficulties in trace analysis such as separation and isolation of substances that are masked by many minor or major mixture constituents, then, in the limits of detection of the chosen technique. Trace components are usually obtained by liquid extraction, distillation, sublimation, ion exchange, chromatography, and coprecipitation among others. Also, the best suited analytical technique must be picked wisely to satisfy the limits of detection. Reagents, water, solvents that are used in sample handling and analyses must be carefully purified to avoid contaminants from the laboratory atmosphere and commonly are called chemically pure, spectroscopically pure, or nuclear pure (over 99.9999% pure) reagents. Also, all used glassware and material must be very clean and used carefully. Trace analysis is becoming of great importance in biochemical and ecological studies, as well as in the chemistry and physics of materials. Errors in trace analysis can be positive or negative. Positive errors are introduced accidentally because of the contamination of samples, while negative errors occur due to the loss of the analyte during sample handling.

Errors in sampling can be random if they are associated with unexpected change, for example, due to environmental conditions like high humidity, wind, or extremely cold weather. On the other hand, an instrumental bias or experimenter accounts for systematic errors that can be reduced to minimal if standardized procedures are employed properly. The choice of solvent is very important when the sample is being prepared based on physical and chemical properties and instrumental requirements for posterior sample handling. For instance, the sample must be soluble in the selected solvent and it is advisable to avoid solvents that are not recommended for the instrumental technique chosen for sample characterization. It is also very important to know an accurate sample concentration. Solid samples are weighed on an analytical balance for greater accuracy and if hygroscopic, samples must be dried before weighing in an oven or desiccator until constant weight. Liquid samples are measured by weight or volume and volumetric flasks are preferred to minimize errors. Alternatively, a glass volumetric pipette can be used, which is calibrated to deliver one accurate volume with the last drop remaining in the pipette. Errors can be compensated using an internal standard, which is an inert substance added in a known amount to the sample before any analysis. This way, random errors become the same for the internal standard and the analyte and the response factor is proportional to the ratio of the analyte and standard concentrations in chromatography or spectroscopy measurements. For systematic errors, such as matrix effects in solution, the ratio will be unaffected as long as the matrix effect is equal for the

standard and the analyte. Although internal standards provide great benefits, it can be difficult to choose one that is suitable. An internal standard must have a signal that is similar to, but not identical to, the analyte. Also, it cannot affect the measurement of the analyte in any way. Finally, the concentration must be well known. This is achieved by ensuring that the internal standard is not natively present in the sample; thus, the only source of it in solution is one that was added to the sample.

Biological samples are collected and placed into pre-cleaned vials (Pawliszyn 2010; Clark et al. 2016; Berndt 2020). Collection vials, flasks, tubes, and analysis equipment must be cleaned according to the procedures or standard methods based on the type of the sample and analysis to be applied. Sample vials must be clean and sterile for handling biological samples and may consist of a specific material that depends on the sample type and the desired analysis. Vials could be made of Teflon, baked glass, polycarbonate, among other materials, and almost all must be cryogenic resistant. Samples must be frozen as soon as possible after collection, shipped to the laboratory on dry ice, and held at a low temperature of −20 °C or −80 °C depending on the sample, until processing (Singh et al. 2020; Mohammedsaleh and Mohammedsaleh 2014).

4 Gaseous Samples in Bioanalytics

Gaseous samples (Lawal et al. 2017) are rare and commonly sampled in environmental and classical analytical areas of research, but there are also some gaseous biological samples, like air (Mochalski et al. 2014) that is exhaled from the lungs, used in the bioanalytical field of research. Nitrogen, oxygen, carbon dioxide, argon, and water are the major constituents of human exhaled breath, whereas as minor components, some volatile organic compounds (Fig. 2) are found and may be useful in diagnostics, and some of those are present just in trace concentrations. Most of the breath volatile organic compounds of interest have low concentrations expressed in parts per million volume (Marco and Grimalt 2015) (ppmv) or lower. Therefore, their reliable detection and quantification (Phillips 1997) show a great challenge not just because it is necessary to use highly sensitive analytical instruments, but also because of the need to concentrate sample (Pijls et al. 2016) prior to analyses. Some of the most useful methods for concentration purposes of the gaseous samples are thermal desorption devices as solid-phase microextraction (Mochalski et al. 2013) (SPME), then, needle trap devices (Amann et al. 2014) (NTDs).

There is also some caution regarding gaseous sample collection. It is first necessary to define the type of breath to be collected, for example, from which phase of exhalation it is: late expiratory, end-tidal, or mixed phase (Das and Pal 2020). Next, whether single or multiple exhalations (Shehada et al. 2014) is going to be applied in sampling. Also, the breath capture (Hakim et al. 2011) technology choice has to be considered as well as the volume of air that is going to pass through the chosen device (200 mL for example). There are some commercially available breaths collection containers used in clinics that have as low as possible ingress of environmental and egress of breath volatiles (Brusselmans et al. 2018). Finally,

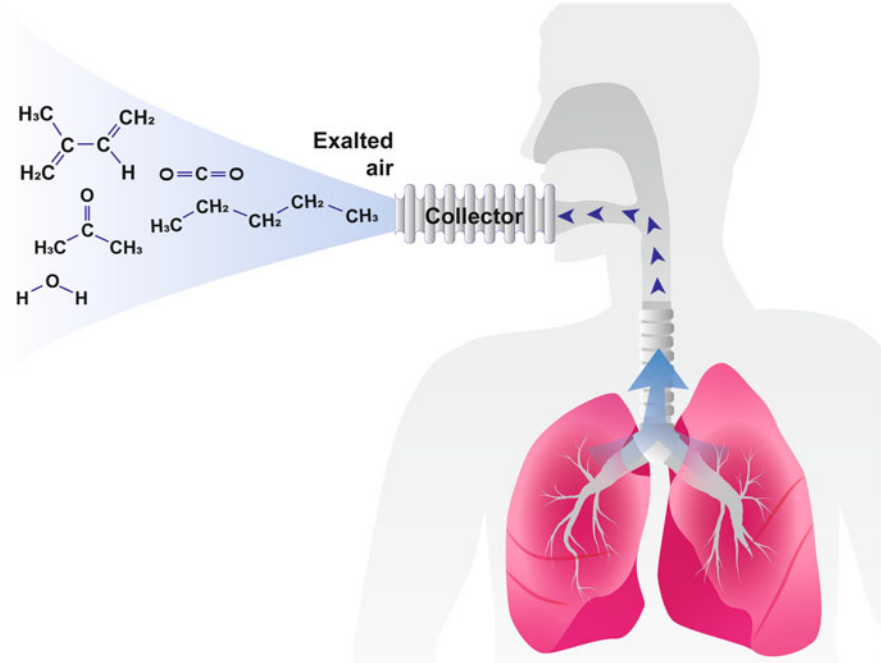


Fig. 2 Gaseous samples in bioanalytics are mostly used in diagnosis such as exalted air for comparative studies. Exalted air contains valuable but low concentration substances like volatile organic compounds (VOCs) that must be collected and concentrated before analysis. Many different types of devices are used to capture VOCs, with filters that explore solid-phase microextraction (SPME), needle trapping devices (NTDs), or others

during storage, some containers may absorb some gaseous sample constituents from the environment that must be considered when data are analyzed as explained as follows (Brusselmans et al. 2018).

It is important to keep in mind that any device prepared for exalted air sampling must be cleaned, usually with high-purity nitrogen (N_2, 99.9999%) in a repeated process (5 times, for example), followed by overnight heating while filled with N_2, and evacuation. After being sampled, exhaled air appears as captured VOCs, which must be desorbed, and analyzed following the supplier instructions to avoid mistakes and flaws in results interpretation. But if needle trapping devices (NTDs) are used, because of their small dimensions and low sorbent masses, there is rapid desorption and direct use in the standard inlets of gas chromatographs. Consequently, contrary to traditional sorbent trapping, no additional equipment, such as a thermal desorber is required. It should also be borne in mind that sometimes it is required to subtract unfiltered air and disregard the identified contaminants in our subsequent analysis when air is taken in hospital conditions.

There is a lot of literature on breathomics (Mochalski et al. 2013; Amann et al. 2014; Das and Pal 2020; Shehada et al. 2014; Hakim et al. 2011; Brusselmans et al. 2018) that is still just a research area with the potential to develop into clinical practice application.

5 Liquid Samples in Bioanalytics

A liquid sample could be obtained from already liquid, biologically relevant samples such as blood and others (Putcha 2000) or from the dissolution of a solid (McDowall 1989). Bioanalysis in clinical chemistry testing, drug, alcohol, and DNA testing and for diagnostic and therapeutic purposes can be applied on many different liquid samples or it can even use more than one type of sample, for example, sweat, tears, saliva, mucous, urine, blood, plasma, serum, breast milk, and cerebrospinal fluid (CSF) for the analysis of the combined results (Nováková and Vlčková 2009). The quantity in terms of the number of samples and volume depends on the type of sample and analysis that is pretended to be applied and also on the availability of a liquid. If researched in clinical testing, the sample size is around 50 mL for blood tests and complete hemogram (Tiwari and Tiwari 2010), but it can be as small as few microliters (Cassiano et al. 2014) if it is difficult to collect, such as in newborns, dangerous to humans on testing or if the analysis demands do not need great sample volumes. The most common handling of a liquid sample is its freezing immediately after the first pre-treatment to very low temperature (Ali et al. 2008) using liquid nitrogen, and then bio-freezer (Liu and Qiu 2020). Also, the sample must be shipped or transported to the laboratory in dry ice and must not be thawed more than one time before sample preparation for suitable analysis (Serra-Mora et al. 2020). Some samples must be analyzed in short periods after sampling, for example, in 2 weeks in the case of the serum samples (Serra-Mora et al. 2020; Mitra 2004; Amiri 2016).

There is a technology that uses an additive in conjunction with centrifuging, which allows human saliva samples (Nasiria et al. 2020) to be stored at room temperature for up to 6 months. Biological liquid samples that require long-term storage at ambient temperatures require preservation prior to analysis. The preservative (Yi et al. 2008) is composed of sodium benzoate in an amount of at least 0.15% of the sample (weight/volume) and citric acid in an amount of at least 0.025% of the sample (weight/volume). The preservative can be dispensed in different platforms such as coated sample collection tubes (Konoz et al. 2016) like vacutainers and other saliva collection devices, coated preservative discs, compressed tablets, or capsules. Larger volume samples are kept in other devices, such as syringes or dropper bottles for preservation.

Often, the various species of interest must be isolated from the original sample prior to analysis. If the liquid sample is a suspension (blood for example), solid particles are removed using filtration or centrifugation before analysis. On the other hand, if it is necessary to isolate analytes, and extraction, separation, or complexation could be used for sample treatment (Amiri 2016; Nasiria et al. 2020; Yi et al. 2008; Konoz et al. 2016). Usually, sample handlings prior to analysis are named pre-treatments and applied to remove any species that may cause interferences, concentrate analytes, or prepare the sample for the chosen type of analysis.

One of the most applied pre-treatments of the liquid samples is extraction by which the target substance is isolated from a solution (Cassiano et al. 2014). Liquid/liquid extraction and solid-phase extraction are some of the explored techniques for isolation of the desired analyte from the complex mixtures where affinity differences

between two phases are crucial for the process. Immiscible liquids are used in liquid/liquid extractions, such as an aqueous phase of the biological samples and an organic phase (chloroform), which form two layers with a distinct boundary (Serra-Mora et al. 2020; Huang et al. 2019). The affinity of the various components within the sample for each of the two layers is used to separate them. So, the analyte distributes between the two different phases according to its properties that englobe partition coefficient that represents the ratio of the solute's concentration in one solvent to its concentration in the second at given temperature and pressure. A very large or very small partition coefficient is considered desirable to enhance extraction success. Also, multiple extractions may have to be performed to remove most of the solute as fractional partitioning of the solute leads to better results, but never to 100% yield (Lin et al. 2020).

Solid-phase extraction is based on the partitioning of an analyte between the sample's liquid and a solid support because of adsorption. Some trace analysis procedures explore solid-phase microextraction (SPME) in sample preparation for the analysis. This technique is sometimes time consuming.

6 Solid and Semisolid Samples

There can be many types of different biological samples that are solids or semisolids, such as hair, tissues, cells, and viruses. Because of that, there are also many different techniques used for their preparation. When samples are in the grind, then obtaining powder from those samples involves grinding a solid sample into a powder using either a mortar and pestle or a ball mill. To ensure that samples are ground and not lost due to sticking to the mechanical apparatus, samples must be dried to expel any water. After reducing particle size and drying, the taken sample has to be homogeneous and representative.

Many analyses require samples that are liquids, so solid samples must be solubilized. Sometimes the sample dissolves easily and is thus ready for analysis or further pretreatment. In other cases, the sample matrix is insoluble in common solvents, and its analytes must be extracted. Analytes may be difficult to remove from the sample insoluble matrix due to adsorption or inclusion phenomena and additional sample preparation steps must be performed to achieve sampling with better accuracy and precision as possible. Besides, smaller particles dissolve faster and are easier to extract because of a greater surface area.

Similar to liquid samples preparation steps before analysis, various extraction methods are used for solid samples, such as solid-liquid extraction, Soxhlet extraction, forced-flow leaching, homogenization, sonication, dissolution, accelerated solvent extraction, supercritical fluid extraction, microwave-assisted extraction, thermal extraction, and others. In solid-liquid extraction, samples are put into closed, or semi-closed containers, and a solvent that can dissolve the targeted analyte, or a desired class of the analytes is added. The solvent is normally kept hot, close to boiling, or refluxed to increase analyte solubilities. After the weight of the extracted

solution becomes stable, the solution is dried, filtered, and then analyzed as it is, or further manipulated to obtain an analyte, or analytes in measurable concentrations.

7 Sampling and Sample Preparation of RNA and DNA

Nucleic acids, RNA, and DNA (Kresse et al. 2018; Farrosh et al. 2011; Lever et al. 2015; Hedegaard et al. 2014) are fundamental biomolecules for the maintenance of life. Their structure is defined by their primary structure that depicts the sequence of purine and pyrimidine bases, starting from their 5′ to 3′ ends, which is determined using sequencing analysis. The success of sequencing analysis depends on the quality of nucleic acid material. DNA samples are more stable compared to RNA, but sometimes nucleic acids isolated from a material kept at low temperatures (−80 °C) for a long might be degraded and difficult to amplify because of degradation or chemical changes that have occurred within the sample (Farrosh et al. 2011). Also, there are many available extraction kits and some efforts have been made to identify optimal RNA and DNA extraction method and kit, nucleic acids quantities, qualities, and the performance of sequencing analysis (Lever et al. 2015). It was observed that for RNA extraction, all protocols gave comparable yields and applicable RNA. But different protocols gave variable quantity and quality of extracted nucleic acids and are to be carefully picked and analyzed prior to any analytical method applied for DNA and RNA characterization (Hedegaard et al. 2014).

Biomedical research samples are in the majority of cases made of tissues that are archived in the form of the formalin-fixed paraffin-embedded (FFPE) blocks. This way, samples from biopsies, surgeries of cancer, and tissues are conserved for long with unchanged morphology (Watanabe et al. 2017; Evers et al. 2011; Howat and Wilson 2014; Williams et al. 1999; Masuda et al. 1999; Head et al. 2014). However, the molecular preservation and quality of the nucleic acids from the FFPE blocks have become critical when discussed the sequencing results. The ability to reliably investigate nucleic acids from FFPE material is thus important, and should also be taken into account in tissue fixation procedures (Gibson et al. 2010). It was evidenced that nucleic acid extraction from FFPE blocks yields a material with an inferior quality compared to one obtained from the fresh frozen tissue (Chory and Pllard 1999). Therefore, determinations of low-frequency genetic variants are very challenging when analyzing only FFPE nucleic acid samples. Nevertheless, the compared data extracted from the paired samples, i.e., fresh frozen and FFPE tissues, showed high concordance in molecular analyses. The nucleic acid can suffer structural changes because of the pH of the fixative, time that spent in the block, the age and storage condition of tissue blocks, and the extraction method used. Because of the chemical reactivity of formaldehyde, which is the active component of formalin, nucleic acids can crosslink with the physically close proteins, or suffer nucleotides' modification (Carini et al. 2016; Eland et al. 2012; Cadillo-Quiroz et al. 2006). Consequently, those factors lead to fragmentation of nucleic acids. In addition, because of the formaldehyde interaction with the key enzymes for sequencing, such as reverse transcriptase, and DNA polymerase among others, can unable

molecular analysis. Furthermore, deamination of cytosine to uracil, and other non-reproducible sequence changes that are introduced, can be misinterpreted as mutations (Eland et al. 2012).

DNA and RNA quantifications, generally referred to as nucleic acid quantification (Esteva-Socias et al. 2020), are usually done to determine the average concentration of DNA or RNA in a sample, before proceeding with downstream experiments, and sample purity is fundamental for an accurate sample analysis. There are two optical technologies generally employed to quantify nucleic acids: UV-Vis and fluorescence measurements. An accurate RNA or DNA quantification is a fundamental step in avoiding posterior experimental errors, and the choice of the most suitable methodology for such a task can save time and money (Srivastava et al. 2020).

It is common to analyze nucleic acids using an electrophoretic run in an agarose (highly purified agar) gel (Fig. 3). The speed of nucleic acid migration depends on ion mobility and size, with the smaller molecules moving faster and appearing at the bottom (end) of the gel. This method is able to separate DNA fragments in the range of 0.1–50 kbp, which is far below the size of bacterial genomes, but includes some naturally occurring genetic elements such as bacterial plasmids. A variant of agarose gel electrophoresis, pulsed-field electrophoresis, alternates the orientation of the electrical field in a fashion that allows resolution of much larger DNA fragments (Lee and Tripathi 2020).

The intrinsic absorptivity of aromatic moieties in nucleic acids, DNA and RNA, is explored for their photometric measurement (Eland et al. 2012; Cadillo-Quiroz et al. 2006). When an absorption spectrum is measured, nucleic acids present a characteristic peak at 260 nm (Fig. 3). It is not selective and does not distinguish DNA from RNA (Gong and Li 2014). It presents limited sensitivity with worst detection limits than fluorescence-based methods. But it is very simple with almost no sample preparation, dispenses use of dyes, or standards, and it can provide purity ratios with direct measurements—$A_{260/280}$ and $A_{260/230}$. It can give information about contaminants, and identify non-nucleic acid contamination in samples (proteins, phenol, guanidine salts), and provide correct concentrations (Bueso et al. 2020).

Another method for DNA or RNA characterization explores fluorogenic dyes that bind selectively with the nucleic acids, and permit their fluorometric measurements. The signal is measured by fluorometers. The sample is excited with filtered light at the excitation wavelength, and the emitted light at the emission wavelength is monitored. This way, characterization of double-stranded DNA (dsDNA), single-stranded DNA (ssDNA), and RNA can be specific-performed. The measurements are sensitive, and nucleic acids in low concentrations, such as pg per mL, can be determined. Therefore, it is the recommended method for samples with very diluted nucleic acid. It is accurate despite contamination being present in the sample, including other nucleic acids as contaminants. The major difference between absorbance measurements and fluorescence measurements in determining the concentrations is that fluorophore is used as an indirect way of detecting molecules like DNA in absorbance measurements. A previously measured standard curve is used to determine sample concentrations (Chory and Pllard 1999; Carini et al. 2016;

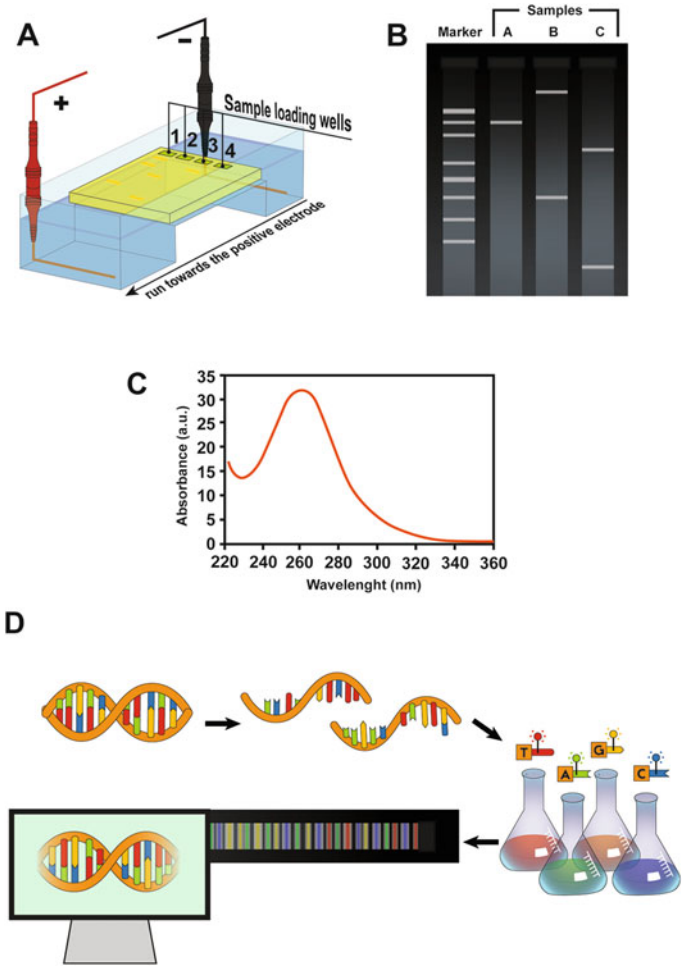


Fig. 3 Illustration of steps used to evaluate the quality of a nucleic acid sample. Nucleic acids are analyzed using electrophoretic runs (**a**) in agarose (0.8–1%) gels, which are analyzed by comparison with standards (**b**) of known molecular masses, and concentration ng mL^{-1} (sizes are expressed in base pairs, bp, with two standards in greater quantity—1000 bp and 500 bp as shown). (**c**) Sample is checked using UV-Vis, and maximum absorbance at 260 nm. (**d**) If clean (highly pure), and obtained in sufficient quantity (ng mL^{-1}), double-stranded DNA (dsDNA) undergoes sequencing

Eland et al. 2012; Cadillo-Quiroz et al. 2006). After combining the standard nucleic acid at least in five different concentrations with the fluorescent dye, a fluorimeter is used to determine the relative fluorescence (RFU).

As already stated, DNA samples can be obtained from different sources such as blood, cells, tissues—frozen, fresh, or stored in paraffin tissue blocks, and so on. DNA extraction and purification involve three steps: (1) breakage of the cells/tissues to expose the DNA, (2) removal of the lipids using a detergent, and

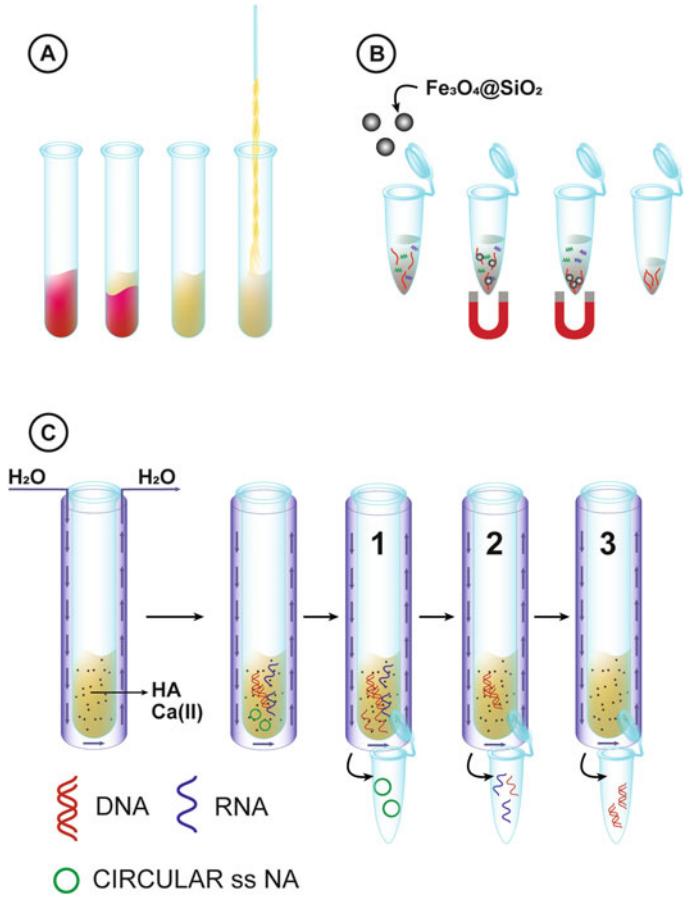


Fig. 4 Illustration of some of the most important isolation methods used for nucleic acid samples (DNA and RNA). (**a**) Classical DNA extraction method: (1) Cells or tissue samples are lysed with detergent and membranes are disrupted, (2) proteases and RNAase are added and lysis of other that DNA occurs, (3) after centrifugation, DNA is separated in the supernatant, and (4) DNA is precipitated with ethanol and can be spooled as viscous stands. (**b**) (1) Magnetic nanoparticles are added to lysed biosamples and bind with nucleic acids, (2) debris is then washed away using a permanent magnet that keeps magnetic beads loaded with the targeted nucleic acids safe at the bottom of the tube or Eppendorf, (3) nucleic acids are rinsed from the magnetic beads and (4) ready to be used. (**c**) Hydroxyapatite (HA) loaded columns are used for different nucleic acids separations such as DNA (single—ss or double-stranded—ds), RNA (ss, ds), and circular nucleic acids using increased phosphate buffer concentrations (1–3)

(3) precipitation of DNA (Head et al. 2014; Gibson et al. 2010; Chory and Pllard 1999; Carini et al. 2016; Eland et al. 2012; Cadillo-Quiroz et al. 2006; Esteva-Socias et al. 2020; Srivastava et al. 2020; Lee and Tripathi 2020; Gong and Li 2014; Bueso et al. 2020). Also, it would be advisable to remove proteins applying proteases, and RNA using RNases (Fig. 4a). DNA purity increases sample stability if aiming for long-term storage.

If one needs to analyze just the RNA, then there are two protocols to be used, isolating and purifying only the RNA from the source material or isolating the total nucleic acids and add DNase to degrade and take off the DNA from the sample or use liquid chromatography for the separation of different nucleic acids. Sometimes, it is needed to extract DNA or RNA simultaneously from many samples, and then it is needed to simplify, speed up and keep less laborious methods for nucleic acid isolation and purification. And for such extractions, magnetic beads or nanoparticles can be used by exploring solid-phase reversible immobilization (SPRI). Magnetic nanoparticles coated with silica ($Fe_3O_4@SiO_2$) reversibly bind nucleic acid under dehydrating conditions and with the aid of a strong magnet, can be safely immobilized and purified through multiple washes, and manipulation steps (Fig. 4b). Binding DNA to silica on magnetic beads is based on altered affinity of the negatively charged DNA backbone to the silica surface in the presence of chaotropic salts such as guanidinium hydrochloride. By varying chaotropic salt concentration, DNA can be selectively isolated (100 and 3000 bp) and recovered up to 95% of the input DNA. Magnetic bead protocols are inherently scalable since they are independent of centrifugation and the required materials are easy to manufacture on a laboratory scale, and not expensive to purchase. Nevertheless, despite many advantages, magnetic materials are not used often.

If just the RNA is needed, then the starting material is disrupted by adding a chaotropic agent (guanidium thiocyanate, for example), and a reducing agent (beta-mercaptoethanol), vigorously shake or vortex, as to break cystine, and inactivate the contaminant proteins. Phenol and chloroform/ *iso*-amylalcohol are added cold to pass RNA into the aqueous phase. Finally, isopropanol is used to precipitate RNA, which is washed with 75% ethanol to remove impurities. RNA is less stable than DNA, so some extra care is needed for RNA sample handling and posterior use in analysis. It is also possible to prepare just the DNA or the RNA sample from the total nucleic acid fraction, which is divided into: (1) the first fraction that after treated with a DNase 1, recovers the RNA, and (2) the second fraction that after treated with RNase, recovers the DNA. Or total nucleic acids can be purified on gel electrophoresis and purified using open column chromatography (Fig. 4c). Hydroxyapatite (HA) has been employed in the separation of nucleic acids for long. HA exploits two primary binding mechanisms: metal affinity interactions through HA calcium—Ca (II), and cation exchange interactions through HA phosphate. Nucleic acids (ssDNA, RNA, and dsDNA) are generally incubated in the HA column (60 °C) and eluted using a phosphate-containing buffer with concentrations that increase (1–3, Fig. 4c). This way, the circular single-stranded nucleic acids are eluted first (the lowest concentration of phosphate ions), followed by linear ssDNA, and RNA (dsRNA and ssRNA), and the last fraction contains just ssDNA (the highest phosphate concentration). As Ca(II) ions in hydroxyapatite can cause differences in applied runs, it is recommended to optimize the elution of total nucleic acids using a previously known standard.

At last, samples containing pure nucleic acids must be properly stored and handled prior to successful downstream research applications. Nucleic acids can degrade at room temperature, especially RNA samples at neutral or low pH from

autolysis, and need to be frozen to maintain their integrity. If the same nucleic acids' sample is going to be used in multiple experiments, it is recommended to make multiple aliquots that will only be thawed once when they are needed. Nucleic acids that are going to be used in a short time frame may be stored at −20 °C but if not, should be kept in an ultra-low freezer, typically at or below −80 °C. Also, an aliquot of a very important sample can be stored in off-site bio-storage (biobank) as backup material to be kept safe and well monitored.

8 Sampling and Sample Preparation of Lipids

Lipids (Granafei et al. 2017) are a very diverse group of molecules that are defined as organic molecules soluble in chloroform. There are many organic compound classes, such as fatty acids and alcohols and their esters, mono-, di-, and triglycerides, sterols, many phospholipids derived from phosphoric acids and esterified with fatty acids, and choline, poly-phosphorylated inositol derivatives, L-serine among others, sphingolipids, and different lipopolysaccharides, which are lipids (Granafei et al. 2017; Abbott et al. 2013; Sajnani and Aribindi 2017; Iverson et al. 2001; Gil et al. 2018). Amphiphilic lipids show dual nature with hydrophilic parts (head group) and lipophilic parts (tails) linked with some polyol (Abbott et al. 2013). Hydrophilic groups could be amino acids, alcohols, amino alcohols, or saccharides. Phospholipids could be mono-, di-, tri- esters, or ethers (Sajnani and Aribindi 2017; Iverson et al. 2001). Even glycerophospholipids with just two fatty acids esterifying the glycerol may show isoforms, with different positions of a given fatty acid on glycerol carbons. Therefore, despite being the smallest biomolecules, lipids are a very complex class of molecules with a considerable number of molecular species that compose lipidome at any given moment. Diverse lipid species and broad application of lipidomics data mean that techniques for isolating the lipid fraction are of considerable interest (Granafei et al. 2017; Abbott et al. 2013; Sajnani and Aribindi 2017; Iverson et al. 2001; Gil et al. 2018; Pizarro et al. 2013; Züllig et al. 2020; Burla et al. 2018; Pellegrino et al. 2014; Bowden et al. 2017).

Lipid isolation processes (Yang et al. 2009; Jiang et al. 2007; Han et al. 2004; Garwolinska et al. 2017; Li et al. 2014) and posterior lipid analyses are of extreme importance for accurate and precise data analyses. For example, if a measured value is very proximate to a standard or known value, the measurement is accurate. On the other side, if two or more measurements are very close to each other, the measurement is precise. Using an example, if a given substance is weighed three times, and gets 2.10008 g each time, then the performed measurement is very precise. If an experiment contains a systematic error, then increasing the sample size increases precision but does not improve accuracy, and the obtained result from inaccurate experiment is just precise. If the systematic error is eliminated, one can improve the accuracy but does not change precision. A measurement system is considered valid if it is accurate and precise (Yang et al. 2009; Han et al. 2004).

The lipophilic fraction of any biological sample comprises all fats, such as triglycerides, fatty acids and alcohols, sterols, waxes, and fat-soluble vitamins

(Garwolinska et al. 2017; Li et al. 2014; Cruz et al. 2016; Correia et al. 2018, 2020; Mor et al. 2020; Satomi et al. 2017). The choice of isolation of this type of lipids must follow the requirements of the desired analyses. Gas chromatography (GC) is the most common technique for identifying fatty acids (FAs). Therefore, lipophilic sample preparation requires isolation of total lipids, separation of the FA compounds from other lipids, followed by transesterification in which the methyl esters of the fatty acids are prepared.

Many different bio-samples can contain lipids, for example, serum, cells, tissues, and viruses (Pellegrino et al. 2014). Also, there are standardized protocols for solid, semi-solid, or liquid samples extraction and handling. Tissues or cells (semi-solid samples), freshly collected or harvested, are the best option for lipid extraction to prevent eventual hydrolysis and oxidation. For example, presence of lysophospholipids, phosphatidic acid, monoacylglycerols, diacylglycerols, or free fatty acids in higher than usual quantities might indicate a possible sample degradation before extraction even started (Bowden et al. 2017; Garwolinska et al. 2017). In some special cases, additional precautions are necessary to minimize rapid degradation of lipid constituents. For example, plant tissues need to be processed immediately after collection by immersion in hot 2-propanol or water (Li et al. 2014). Plant phospholipase D is known to be active even in some solvents. If tissue cannot be immediately handled for extraction in a laboratory (off-site sample collection), then, it should be frozen as quickly as possible in a dry ice or better in a liquid nitrogen and stored in sealed glass containers at −70 °C (Cruz et al. 2016; Correia et al. 2020).

Fresh or frozen tissues or cells should be homogenized in the chosen solvent mixture and agitated for a fixed time before elimination of the solid part. Soft tissues may be minced with scissors in the cold and homogenized solvent with an Ultraturax type device, a Waring blender, or a glass-Teflon Potter. Hard tissues (muscle, vessels, bone among others) are best pulverized in a mortar filled with liquid nitrogen. Cells might be treated directly with the solvents (Bowden et al. 2017; Yang et al. 2009; Jiang et al. 2007; Han et al. 2004; Garwolinska et al. 2017; Li et al. 2014; Cruz et al. 2016; Correia et al. 2018, 2020; Mor et al. 2020; Satomi et al. 2017).

The classical Bligh-Dyer method uses chloroform/ methanol/ water solvent mixture but there are some chloroform-free solvent extraction protocols based on 2-methoxy-2-methylpropane also known as methyl *tert*-butyl ether (MTBE) protocol (Bowden et al. 2017; Garwolinska et al. 2017). Replacing chloroform with MTBE is safer, somewhat less carcinogenic and toxic, with lipids dissolved in the upper solvent phase, which allows an easier pipetting, and has the potential for automation (Correia et al. 2020).

It is important to have in mind that lipids must not be stored in dry form for at least two reasons. Oxidative degradation is slower in solutions even when antioxidants are not added. Air and light influences must be diminished. The best storage conditions are bio-freezers (−70 °C) for a long storage (up to 1 year) of lipid extracts in chloroform–methanol mixtures that fill well-sealed stopper glass vials (with Teflon liners), the cap is secured with a wide length of self-sticking tape. During long periods of storage, it can be useful to flush vials or tubes with nitrogen

before closing to prevent fatty acid oxidation. Also, for the same purpose, a low number of antioxidants, such as butylated hydroxytoluene (BHT) or 2,6-di-tert-butyl-4-methoxyphenol or ethyl gallate (around 50μg BHT/mL), may be added to the solvent, principally because the purified lipid extracts usually contain low quantities of natural antioxidants. The added antioxidant is commonly prepared as a concentrated solution in ethanol (10 mg mL^{-1}) (Han et al. 2004; Li et al. 2014; Correia et al. 2018).

Some lipid samples show decrease in esters and plasmalogens, and an increase in free fatty acids, diacylglycerols methyl, or ethyl esters, when stored for a long time (Satomi et al. 2017). These problems can be minimized by using neutral extracts and 2-propanol instead of methanol or ethanol in the solvent mixture. It must be remembered that the best results are obtained with freshly prepared materials.

9 Sampling and Sample Preparation of Carbohydrates

Carbohydrates (Sanz and Martínez-Castro 2007) are polar molecules that occur as free, such as mono-, oligo-, and polysaccharides, or as covalently bonded to other biomolecules in the form of bioconjugates, with lipids (glycolipids) or proteins (glycoproteins and proteoglycans) (Sanz and Martínez-Castro 2007; Montañés et al. 2008; Soria et al. 2012a; Melini and Melini 2020; Aoki-Kinoshita et al. 2020). In general, carbohydrates are hygroscopic, but polysaccharides are mostly stable in the solid state, at physiological pH and standard temperature, but labile thermally as they can decompose losing water. Carbohydrates are also grouped into structural and non-structural, and the last ones are abundant in plants (Sanz and Martínez-Castro 2007). If plant material is collected for analysis, then, it is important to avoid sampling degradation effects by diminishing any stress caused (cutting), and inhibit enzymes, using shock-freezing, transportation in coolers (ice or dry ice) to the laboratories, and posterior freeze or oven drying (Ou et al. 2018). Nevertheless, the most important steps to the successful analyses of carbohydrates are not just sampling, and sample handling, but sample preparation, such as the extraction protocol and the analytical methodology used for soluble sugar quantification, and characterization (Melini and Melini 2020; Ou et al. 2018).

There is a fairly large literature on carbohydrates' analysis in food, and many articles and methods have been written or introduced since glycoconjugates have been the objects of study (Soria et al. 2012a; Melini and Melini 2020; A successful analysis of carbohydrates starts with the first step, which is to pick a representative sample, then, proceeds with suitable extraction, and finally, carbohydrates are analyzed.. Aoki-Kinoshita et al. 2020). Also, sample handling and preparation are different if water- or alcohol-soluble carbohydrates, such as mono- or oligosaccharides or polysaccharides, or glycoconjugates are to be analyzed (Chlumská et al. 2014; Landhäusser et al. 2018; Olajos et al. 2008). If the molecules are freely soluble, then the sample preparation is relatively simple, but, still, other biomolecules need to be removed.

Mono- and oligosaccharides are soluble in alcoholic solutions, thus, the semisolid or solid samples, after drying to the constant weight, are treated with 80% ethanol to yield extracts free from proteins and polysaccharides (Melini and Melini 2020; Khatri et al. 2017; Lu et al. 2018; York et al. 2020; Yamada et al. 2020). Such way prepared extracts are still not pure enough for posterior analyses because of other ethanol-soluble molecules—organic and amino acids, vitamins, minerals, pigments from the original sample, and further purification is achieved by clarifying, and use of different chromatography techniques (Chlumská et al. 2014; Landhäusser et al. 2018; Olajos et al. 2008; Khatri et al. 2017; Lu et al. 2018).

For example, thin-layer (TLC), and liquid chromatography (LC) are commonly used to isolate carbohydrates (Melini and Melini 2020; Lu et al. 2018; York et al. 2020; Yamada et al. 2020). Carbohydrates are separated according to their intermolecular interactions with a stationary and a mobile phase, and isolated as bands in TLC and fractions in LC (York et al. 2020; Yamada et al. 2020; Jensen et al. 2012; Trbojević Akmačić et al. 2015; Fomin et al. 2020; Bertagnolli et al. 2014). Carbohydrates can be separated by electrophoresis but only when previously modified into charged species, which after being applied to a gel, show different ion mobility in an electric field (Soria et al. 2012b; Mort and Pierce 2002). Thus, migrate with different speed and are separated based on their size (Dvořáčková et al. 2014), and smaller molecules move faster.

There is no universal procedure for the analysis of most polysaccharides. Therefore, after isolation, and hydrolysis, their constituent monosaccharides are determined (Chlumská et al. 2014; Khatri et al. 2017). Enzymic hydrolysis methods are specific and sensitive but not selective, break up all glycosidic bonds with the specific type—alpha or beta, and used rarely. Dietary fibers, soluble, and total fibers, are composed primarily from non-starch polysaccharides, and analyzed after the removal of the digestible starch with treatment with amylases, proteins are removed with protease, and a nondigestible residue is obtained for further analyses (Melini and Melini 2020).

Sampling and sample pretreatment are very important stages that precede carbohydrate analysis, and depend on the type and composition of the analyzed sample (Soria et al. 2012a; Melini and Melini 2020; Aoki-Kinoshita et al. 2020; Ou et al. 2018; Chlumská et al. 2014; Landhäusser et al. 2018). Sample preparation includes filtration, dialysis, extraction, reverse osmosis, chromatographic fractionation, and biological and chemical treatments. As already explained, the first step is the separation of carbohydrates from the others components, such as lipids and proteins. Ultrafiltration and filtration using membranes are used for sample concentration. Complex biological samples with different classes of carbohydrates in various amounts require the use of preparation techniques before analysis.

Carbohydrates can be extracted using common techniques, such as liquid-liquid and solid-phase extractions (SPE), where the sample affinities between the stationary and mobile phases are explored (Sanz and Martínez-Castro 2007; Melini and Melini 2020). Columns such as C8, C18, porous graphitic carbon, strong cation exchangers are some of the most used as a stationary phase. For instance, mono- and oligosaccharides from many types of honey can be isolated by active carbon and

celite columns and ethanol/water as a mobile phase. Nevertheless, the successful results are obtained with ease in LC, the high quantities of solvents, and low analyte concentrations are bottlenecks of these sample preparation methods. Therefore, other methods might be a better choice.

One of such methods relies on extraction with supercritical fluids (SFE) that uses pure fluid or a mixture, such as fluid with the additives, under critical pressure and temperature (Montañés et al. 2008; Melini and Melini 2020). This method explores carbohydrate solubility in a chosen liquid fluid, and diffusivity in the gas under supercritical conditions. Because of the moderate critical temperature and pressure that CO_2 has, it is the most commonly used fluid in SFE. Extracts become free of solvent because CO_2 turns gaseous under ambient conditions. Finally, the selectivity of the extraction can be controlled with the depressurization steps. However, CO_2 has a low polarity, which makes the solubilization of carbohydrates difficult, and different amounts of organic solvents are employed as CO_2 modifiers.

Sometimes, it is recommended to apply pressurized liquid extraction (PLE) for carbohydrate sample preparation (Melini and Melini 2020). This method explores just the solvent liquid state at high pressures. Thus, in a short time, it can extract high carbohydrate amounts using lower solvent volumes. PLE is mostly used to extract carbohydrates from food and natural products, and counts on application of various solvents such as water, organic solvents, and water/organic solvents mixtures (Melini and Melini 2020). One interesting example is purification of lactulose from mixtures with lactose using PLE with 70:30 (v/v) ethanol/water mixture (Melini and Melini 2020). Another example is the isolation of di- and trisaccharides from honey by using PLE combined with the use of activated charcoal (Melini and Melini 2020).

Another method for isolating carbohydrates that explores different mobility of the mixture components is field-flow fractionation (FFF) (Melini and Melini 2020). The solution of a mixture of carbohydrates is fractionated when pumped in a laminar flow that suffers a perpendicular effect, which can be gravitational, thermal, magnetic, or electric. Because of the differences in the mobility, components are separated. For example, polysaccharides such as starch, cellulose, pullulan, among others, are commonly isolated from smaller carbohydrates this way (Melini and Melini 2020).

The size exclusion chromatography (SEC) is used to separate oligo- and polysaccharides and elucidate the molecular mass distribution of carbohydrate samples. Another important technique explores ion exchange on anion or cation exchange resins for oligo- and monosaccharides extraction (Sanz and Martínez-Castro 2007; Melini and Melini 2020). On the other hand for large-scale palatinose- and trehalose fractionation, the simulated moving bed chromatography (SMBC) is used (Montañés et al. 2008). Based on adsorbents, ligand-exchange chromatography or size-exclusion chromatography are also employed to raffinate and extract samples for posterior analyses. For example, lactose from a complex sample of human milk oligosaccharides is purified using SMBC (Melini and Melini 2020).

Carbohydrates can also be separated according to their molecular masses exploring dialysis, ultra-, and nanofiltration systems, and different membranes. A sample that contains different carbohydrates can be fractioned using adsorption onto an

activated charcoal column, followed by a subsequent elution with ethanol-containing solvents. This way, with the lowest concentration of ethanol (1%) monosaccharides are the first fraction, disaccharides need more concentrated ethanol (5%), and oligosaccharides are obtained just after using the highest concentrated ethanol solution (50%) (Melini and Melini 2020). This is one of the best choices to isolate monosaccharides from a very complex sample. Other sample preparation methods employed in carbohydrate analysis include the use of molecularly imprinted polymers (MIPs). Such MIPs have been demonstrated to be useful in the recognition of different saccharides or the epimeric differentiation of disaccharides.

Different chemical procedures are used to prepare carbohydrates for analysis, starting from their hydrolysis, followed by low molecular mass mono- or oligosaccharides analyses. Poly- and oligosaccharides hydrolysis treatment with strong acid (trifluoroacetic and hydrochloric acid, sometimes, sulfuric acid) whose concentration, as well as temperature, and reaction time must be optimized to complete hydrolysis and avoid sample degradation (dehydration). Sometimes, polysaccharides are just partially hydrolyzed to mono- and oligosaccharides whose analysis might be useful to determine the polysaccharide structure (Olajos et al. 2008; York et al. 2020).

Carbohydrates have very similar polarity, show low volatility, and do not have chromophore (Yamada et al. 2020). Therefore, if wished to be analyzed by GC, carbohydrates should be derivatized to increase the volatility. Monosaccharides are transformed into alditol acetates by reduction and acetylation, while oligosaccharides become volatile if turned into trimethylsilyl or trifluoromethyl ethers. If meant to be detected after a run in HPLC, because carbohydrates can only absorb UV light below 195 nm, amperometric or refractive index detectors must be used. On the other hand, it is advisable to modify their structure with UV absorbing or fluorescent labels with derivatization processes to increase the sensitivity of their detection. The reaction, which is the most usual procedure pre-column derivatization, may take place before or during the analysis (Jensen et al. 2012; Trbojević Akmačić et al. 2015). The most employed approach needs a reductive amination. These procedures are based on the condensation of a carbonyl group with primary amines to produce a Schiff base, which is later on reduced to give an *N*-substituted glycosyl amine. 2-Aminopyridine trisulfonates are among the chemicals more employed. Besides, considering that most carbohydrates are neutral and hydrophilic compounds, derivatization produces not only a sensitivity enhancement but also polarity and electrical charge variations, which can facilitate their separation afterward (Fomin et al. 2020).

At last, the carbohydrate field of research called glycomics is becoming popular and of great interest. Glycomics is a study that maps all carbohydrates of an organism, which may be free or present in bioconjugates, such as glycolipids, glycoproteins, and proteoglycans, and links any deviation from a healthy state to provoked with inherited, and genetic reasons, physiological changes, pathological, and other problems. If analyses of carbohydrate constituents of glycoproteins are in question, then, 50–100 $\times$ 10^{-6} g of pure protein in solution is needed. After the

release of the *N*-glycans, labeling and purification steps are held, the sample is dried and resuspended into 20μL and analyzed just 2μL. If a glycoprotein band is analyzed after SDS gel electrophoresis, around 10μg of the sample is needed (Bertagnolli et al. 2014; Soria et al. 2012b; Mort and Pierce 2002; Dvořáčková et al. 2014; Szpakowska et al. 2020; Deng et al. 2004).

10 Sampling and Sample Preparation of Proteins

Chemically speaking, proteins (Matejtschuk 2007) are biopolymers made from amino acids linked through peptide bonds in one or more chains of polyamides (polypeptides), which start at the N-terminal and end at the C-terminal. Proteins show a defined number of monomers per chain, high molecular masses expressed in kDa, and characteristic isoelectric points (pI) (Feist and Hummon 2015). Some proteins contain other chemical entities called prosthetic groups that are important for their functions such as heme in hemoglobins, cofactors (e.g., NAD^+) in enzymes, among others (Feist and Hummon 2015; Gutstein et al. 2008; Aebersold and Mann 2003). Other proteins appear as bioconjugates like glycoproteins and lipoproteins (Gutstein et al. 2008). They may be functional just if dimeric (insulin receptor, for example) or multimeric (ATPase), and make complex homo- or heteromeric structures. Proteins are important structural components and also have multiple and very important functions as enzymes, as well as take part in cell signalization, transport, DNA replication, and translation, and are molecules with primordial roles that orchestrate life (Feist and Hummon 2015; Gutstein et al. 2008; Aebersold and Mann 2003; Medzihradszky and Chalkley 2015; Scott et al. 2012). They are very diverse and can be rigid and elastic as keratin and silk, respectively, and very sensitive as even a small change in pH, ionic force, and temperature might drastically change their structure. And as all biomolecules, proteins are found in complex mixtures and diverse physical states and can be present in traces or as major components (Feist and Hummon 2015; Gutstein et al. 2008; Aebersold and Mann 2003). Therefore, likewise in other sample handlings, some care must be taken when analyzing proteins, and many of them depend on our target analysis if one wishes to determine and describe all proteins present in a sample (proteomics), or analyze the structure or function of one targeted protein, thus, two different approaches in sample preparation are undertaken (Gutstein et al. 2008; Scott et al. 2012).

If our objective is proteomics, a study of the total protein content of a cell, a tissue, an organ, or a whole organism at a given moment, then proteins must be isolated from other mixture/sample components and analyzed as such. The proteomic sample usually contains hundreds or thousands of proteins in very diverse concentrations. For example, human blood serum is made from 60 to 80 mg/mL of proteins (Hilbrig and Freitag 2003; Doellinger et al. 2020), whose half accounts for human serum albumin (HSA) and almost ¼ to γ-globulins, while others are present in traces, such as cytokines. Thus, even highly sensitive, analytical procedures based on mass spectrometry (MS) or electrophoresis are not capable of detecting all the protein species present in the sample (Hilbrig and Freitag 2003). Without a targeted

approach to improving detectability, the low-abundance part of proteomes is in essence inaccessible or difficult to detect (Wiśniewski 2019).

Even if it is our aim to characterize all detectable proteins in a sample, there is a need for sample preparation after collection. The quality of such preparation is critical to successful protein analysis to ensure the best possible analytical results. There are many methods for preparing protein samples for MS analysis. Before deciding on the best method for obtaining the sample, there is some information needed, such as the type, biophysical properties, and quantity of the targeted proteins, their location, and complexity of the samples (Wiśniewski 2019; González-García et al. 2020; Kachuk et al. 2015; Eddhif et al. 2018). Preparation protocols that employ lysis, precipitation, and fractionation of proteins, depletion of high-abundance proteins, then enrichment of selected proteins, and mass tagging tools to accurately identify and quantify the analyzed samples are often used (González-García et al. 2020).

The use of chromatographic systems is a part of sample preparation if aiming to protein fractionation (Visser et al. 2005) in proteomics studies, and sometimes it is applied to isolate and characterize just the targeted protein (Bache et al. 2018). If aiming to study the pure protein and characterize it, then it is necessary to analyze the mixture and determine where the target protein is by applying an electrophoretic run, to evaluate the target protein concentration and its similarity in size with other mixture proteins and choose the best way for posterior purification protocol. The most efficient purification procedure employs affinity liquid chromatography (Fig. 5), usually counting on antibody–-antigen or lectin–glycoside interactions, but it is difficult to perform it if the targeted protein is not previously known (Müller et al. 2020). There are many different resins, magnetic and magnetic agarose beads commercially available among which, the ones for the immunoprecipitation (IP) techniques are the most efficient. For example, protein G magnetic beads for immunoglobulins purification from human samples can be used to purify up to 25–30μg human IgG/mg beads (Doellinger et al. 2020; Vuckovic 2013; Zeki et al. 2020; Wawrzyniak et al. 2018). After each purification step, it is necessary to evaluate how efficient was the purification, and electrophoretic runs must be performed. For example, Fig. 5 shows an illustration of the purification of five proteins whose molecular mass can be estimated by comparison of the run that molecular mass marker (M) showed under the same circumstances. Also, before any biophysics method is used for the target protein characterization, its concentration must be determined usually using UV-Vis protein properties (280 nm) or other colorimetric methods based on the application of dyes, like the Bradford method (Coomassie) or fluorescent dyes with many kits sold commercially.

Proteins can be analyzed using electrophoretic runs in 1D and 2D where differences in the molecular masses and isoelectric points are applied to separate proteins. Digest-in gel trypsin lysis for proteomics research is one of the ways to prepare proteins for MS and if the targeted protein exists in a database, only 10–20 ng of the protein is needed. For the identification of proteins in complex mixtures (e.g., separated over a gel lane), the amount of protein should be around or greater than 10μg. For the protein molecular mass determination, at least 10μg at a

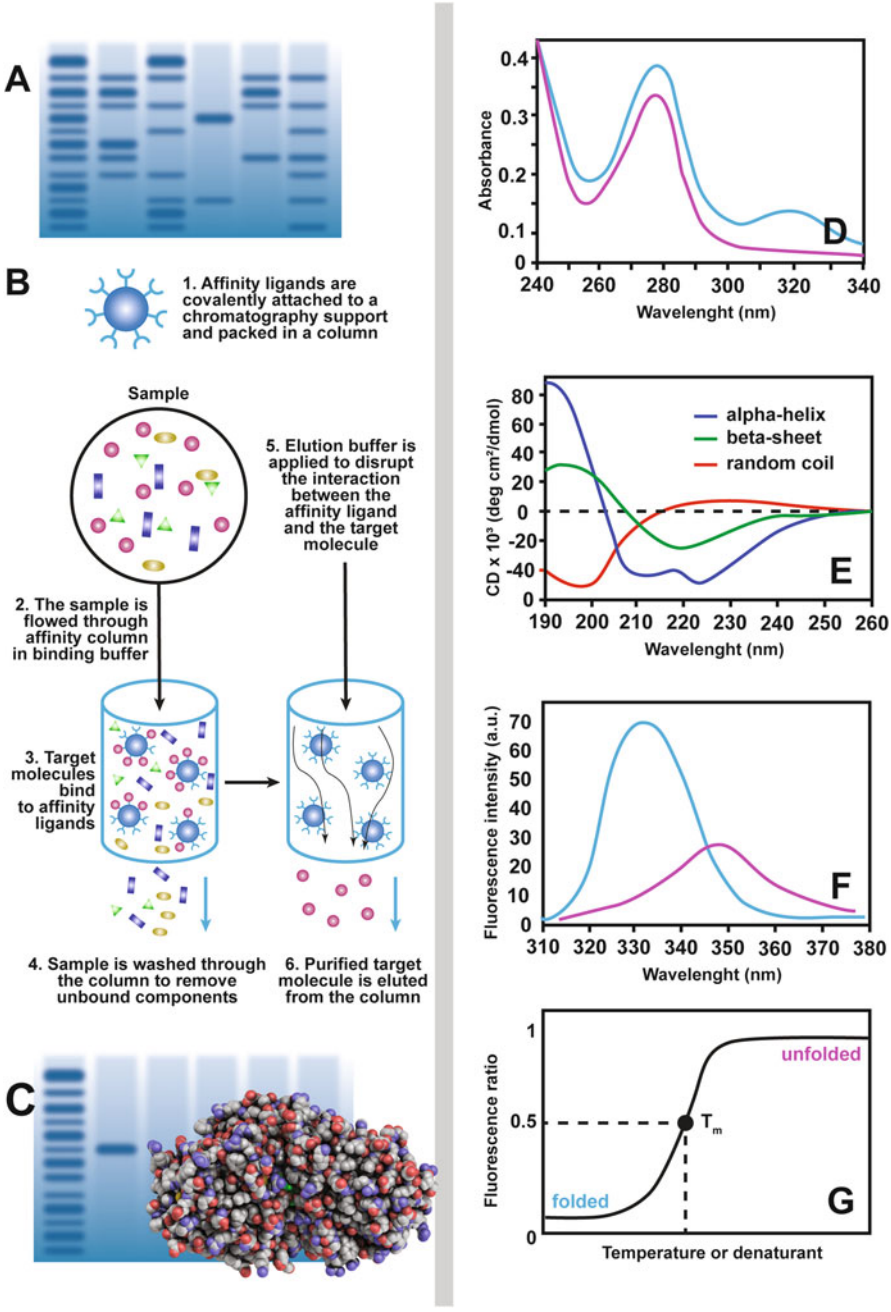


Fig. 5 Illustration of principle steps needed for the target protein analysis. The target protein is isolated from a complex mixture by precipitation of total proteins. After an electrophoretic run to evaluate concentration and size of the target protein, the best strategy for its purification is proposed,

concentration not lower than 1 mg mL^{-1} is needed (Hughes et al. 2019; Wiśniewski 2019; González-García et al. 2020). There are some important tips to keep in mind, the Coomassie-staining of the gel cannot be done in a microwave or scanned on an overhead projector foil, because microwaving modifies proteins in the gel, and there is no way to extract them, also polymers in the overhead projector foils cause trypsin inhibition (Kachuk et al. 2015).

Proteins in a given sample can be characterized at the proteome or sub-proteome level (Kachuk et al. 2015; Eddhif et al. 2018), for example, one can be interested in determining just the cell phosphoproteome to understand their role and function. The big issue in proteomics studies is linked to separation methods that precede MS analysis. There are some technologies that show great potential for undirected and targeted MS. For example, ion mobility separations, microchip-based, chromatographic nanoscale reversed-phase liquid chromatography, capillary electrophoresis are just some techniques that can be hyphenated with MS.

Another important issue to be considered is protecting proteins from the action of proteases, and other protein-modifying enzymes. It is very important to preserve the real state of the proteins and proteomes from different tissue and cell samples so that analytical results can be trusted (Hughes et al. 2019; Wiśniewski 2019; González-García et al. 2020; Kachuk et al. 2015). If not controlled, degradation and protein modification can potentially take place throughout the complete sample preparation workflow (Eddhif et al. 2018; Bache et al. 2018).

Similar to some biological samples, proteins are sensitive to rough handling, especially if concentrated and purified (Müller et al. 2020). Proteins can degrade during storage due to breakdown, aggregation, and inactivation (Nickerson and Doucette 2020). And while there are many storage options, each has its own tradeoffs. If used briefly or within 2 weeks, concentrated protein samples (1 mg mL^{-1}) should be kept in clean, sterilized glassware, or polypropylene tubes at 4 °C or stored in buffers with the appropriate protease inhibitors, antimicrobials, and reducing agents, respectively (Doellinger et al. 2020). In other circumstances, it is recommended to keep protein samples in the freezer, −20 °C with the cryoprotective agents (Vuckovic 2013).

Fig. 5 (continued) being affinity-based bioanalytical method as one of the most efficient if the very pure and low quantity of protein is needed for further analyses. Protein concentration is estimated using UV-Vis measurements, using aromatic acid residues and cystine as protein chromophores or dyes (Coomassie or fluorescent dyes). Biophysical methods that are mostly used to evaluate protein 3D and 2D features are fluorescence and circular dichroism. As proteins exhibit their function only when found in native conformation, their stability (thermal, solvent, like pH and ionic force) might be analyzed by either technique

11 Small Molecules Samples

The smallest molecule is the hydrogen molecule (H_2), but the term small molecule in bioanalytics refers to molecules found in biological samples that show molecular masses lower than 1.5 kDa (Zeki et al. 2020). There are many powerful analytical techniques and hyphenated ones that are frequently employed in attempts to characterize as many as possible small molecules or metabolites within a given biological sample (metabolomics). Some of those need little or no preparation of samples and others need extensive sample preparation and standardized procedures (Zeki et al. 2020; Wawrzyniak et al. 2018; Roberts et al. 2012; Lin et al. 2007; Wenk 2005; Auslander et al. 2016; Wolthuis et al. 2020; Chong et al. 2018; Coley et al. 2019; Pontes et al. 2016; Pontes et al. 2017). Differently from biomolecules cited in previous chapter sections, small molecule sample preparation is not based on isolation and purification of all of those present in the matrix to be analyzed, but in getting great samples from small molecules based on their polarities. Usually, metabolites are split into polar (Wawrzyniak et al. 2018) and nonpolar (Roberts et al. 2012) isolates and further fractioned (Lin et al. 2007) if needed. In general, those methods are long, require the use of chemicals, such as solvents for extraction, isolation, and purification, other consumables, for example, for water sorption, and prone to errors because of human engagement.

There are two metabolomics approaches, targeted and untargeted (Lin et al. 2007), which are used in biological fluids and tissues analyses. Sample preparation methods for the untargeted metabolomics are different if nuclear magnetic resonance or mass spectrometry as analyzers are applied afterward. It is advisable to keep both sample preparation and handling simple to avoid errors, such as contamination and sample degradation, and quench any sample decomposition to preserve intact metabolome composition as at the time of sampling (Wenk 2005). There are also in vivo methods such as microdialysis (Zeki et al. 2020; Wolthuis et al. 2020), solid-phase microextraction, turbulent flow chromatography, and dried blood spot sampling. Among various sample preparation methods, it is advisable to pick and test the most appropriate one for a given cohort of samples.

For example, in untargeted metabolomics by LC-MS, sample preparation methods include protein precipitation, liquid-liquid extraction, solid-phase extraction (SPE), or ultrafiltration. Protein fraction must be removed using organic solvents or a mixture thereof, followed by centrifugation and supernatant filtration, which are the most frequently used approaches in plasma untargeted metabolomics. The success of extraction in untargeted metabolomics depends on extracting the widest possible range of metabolites from the sample. In targeted metabolomics, the protocol employed for extraction must be efficient for getting just the class of desired metabolites or one metabolite. Thus, tailored or personalized protocols need to be developed.

Independently on the approach applied in metabolomics study, biological replicates and confounders are to be kept in mind when sampling and sample preparation are in question. For example, if human samples are analyzed, investigated individuals must be matched when considering gender, age, health

problems, because of the biological variability, and over 50 individuals in each group are recommendable but not limited, and the more the better. If used animal models, then age (weeks), type, diet, exercises must be identical among the studied groups, and at least five plus samples must be investigated. When samples are wild species, then at least five specimens are needed for good statistics that englobe data analysis (Roberts et al. 2012; Lin et al. 2007; Wenk 2005; Auslander et al. 2016; Wolthuis et al. 2020; Chong et al. 2018; Coley et al. 2019; Pontes et al. 2016, 2017).

Alongside the text and in many items, it was pointed that biological samples must be frozen just after being sampled and before transportation to a laboratory, kept at low temperatures and that most of the samples cannot be thawed for more than one time to protect them from degradation or modification of their composition (Lin et al. 2007). Some tissues and cell samples and even biomolecules need to be kept together with the cryoprotectant agents, which are used to prevent ice formation, which causes freezing damage (Lin et al. 2007). Cryoprotectants reduce the ice formation at any temperature by increasing the total concentration of all the solutes present in the system by lowering the melting point of water (Auslander et al. 2016). In order to be used, cryoprotectants must have the capacity to penetrate the cells and show no or very low toxicity. For example, one of the most used cryoprotectants is ethylene glycol, although dimethyl sulfoxide (DMSO), and glycerol are used sometimes (Auslander et al. 2016). Usually, 5–15% of cryoprotectants is enough to allow the isolated cells to survive after freezing and thawing from the liquid nitrogen temperature (Chong et al. 2018; Coley et al. 2019; Pontes et al. 2016; Pontes et al. 2017).

12 Final Remarks and Future Perspectives

Sample preparation is one of the most important features in bioanalytical chemistry, and most of the samples are sampled or collected, kept, and transported to the chemical laboratory using standardized procedures and techniques. The first decision that is the key for successful chemical analysis is how to handle and keep the samples to maintain them with the conserved composition until the data are not analyzed and published. If chemical bioanalysis goals are trace molecules and their quantification, then sample handling may need extraction, purification, and concentration techniques and usually, biological samples need the most demanding preparation methods. The laborious methods that need a lot of time to complete are prone to errors and thus laboratory automation can be expected to have a substantial impact on the quality of the sample preparation. Also, improvements in instrumentation and detection limits for many substances may lead to the miniaturization of extraction and purification systems that reduce quantities of samples and chemicals. There are many micro-methods with new materials used as sorbents, micro-devices that may substitute the macro-scale sample preparation. Also, in recent bioanalytics, there are many nanomaterials-based protocols for biomolecules isolation, purification, and concentration. Many employ single-drop micro-extraction, tailored magnetic nanoparticles (magnetic beads) because of their easy separation, molecularly

imprinted nanoparticles, polymer-coated nanoclusters, and others. Purification protocols based on nanomaterials explore electrostatic and hydrophobic interactions, but also other specific interactions such as mimetic enzyme–substrate or receptor–ligand interactions, or metal affinity interactions, which may reduce the time needed for sample preparation. Therefore, the development of new nanomaterials to improve the effectiveness of the sample preparation is a new and promising field of research. Some of the proposed protocols still need to be adopted for day-to-day research activities and use, but certainly will enhance sensitivity and selectivity in sample handling. And because they are easy and fast, even if not cheap, they are becoming interesting to apply due to their effectiveness in the extraction, enrichment, and purification of targeted compounds. Trends in this area of science are to use as little as possible of a starting sample, get as much as possible chemical data on it, characterize the sample composition in a short time in an optimized and rather automated way.

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